Exercise C - II

SUBJECTIVE / INTEGER TYPE QUESTIONS

1. If $\sqrt{x^2+y^2} = e^{\arcsin\frac{y}{\sqrt{x^2+y^2}}}$. Prove that $\frac{d^2y}{dx^2} = \frac{2(x^2+y^2)}{(x-y)^3}$, $x > 0$.

यदि $\sqrt{x^2+y^2} = e^{\arcsin\frac{y}{\sqrt{x^2+y^2}}}$. तो सिद्ध कीजिये $\frac{d^2y}{dx^2} = \frac{2(x^2+y^2)}{(x-y)^3}$, $x > 0$.

Ans.

Sol. $\sqrt{x^2+y^2} = e^{\sin^{-1}\left(\frac{y}{\sqrt{x^2+y^2}}\right)}$

$$\Rightarrow \frac{1}{2}\ell n\left(x^2+y^2\right) = \sin^{-1}\left(\frac{\frac{y}{x}}{\sqrt{1+\left(\frac{y}{x}\right)^2}}\right)$$

$$\Rightarrow \frac{1}{2}\ell n\left(x^2+y^2\right) = \tan^{-1}\left(\frac{y}{x}\right)$$

Differentiating with respect to 'x'.

$$\Rightarrow \frac{x+y\frac{dy}{dx}}{x^2+y^2} = \frac{x\frac{dy}{dx}-y}{x^2}\cdot\frac{x^2}{y^2+x^2}$$

$$\Rightarrow x+y\frac{dy}{dx} = x\frac{dy}{dx}-y$$

$$\Rightarrow (y-x)\frac{dy}{dx} = -(y+x)$$

$$\frac{dy}{dx} = \frac{y+x}{x-y}$$

Again differentiating with respect to x

$$\frac{d^2y}{dx^2}=\frac{(x-y)2-2x\left(1-\frac{dy}{dx}\right)}{(x-y)^2}$$

$$=\frac{2}{(x-y)^2}\left[X-y-x\left(\frac{X-y-x-y}{X-y}\right)\right]$$

$$=\frac{2}{(X-y)^3}\left[(X-y)^2+2xy\right]$$

$$=\frac{2\left(X^2+y^2\right)}{(X-y)^3}$$

2. Find a polynomial function $f(x)$ such that $f(2x)=f'(x)f''(x)$.

एक ऐसा बहुपद $f(x)$ बताइए जिसके लिये $f(2x)=f'(x)f''(x)$ है।

Ans. $\frac{4x^3}{9}$

Sol. Let $f(x)=a_0x^n+a_1x^{n-1}+...+a_n$

Given f(2x) = f′(x) f′′(x) _____(1)

↓ ↓ ↓

degree degree degree

n n – 1 n – 2

both side highest power

$n=n-1+n-2$

$n=3$

$\Rightarrow f(x)=a_0x^3+a_1x^2+a_2x+a_3$

$\Rightarrow$ from (2)

$8a_0x^3+4a_1x^2+2a_2x+a^3$

$=(3a_0x^2+2a_1x+a_2)(6a_0x+2a_1)$

comparing coefficients of equal power

$8a_0 = 18a_0^2 \Rightarrow a_0 = \frac{4}{9}$

$a_1 = 0 \qquad a_2 = 0 \qquad a_3 = 0$

$f(x) = \frac{4}{9}x^3$

3. If $2x = y^{\frac{1}{5}} + y^{-\frac{1}{5}}$ then $(x^2-1)\frac{d^2y}{dx^2} + x\frac{dy}{dx} = ky$, then find the value of 'k'.

यदि $2x = y^{\frac{1}{5}} + y^{-\frac{1}{5}}$ तो $(x^2-1)\frac{d^2y}{dx^2} + x\frac{dy}{dx} = ky$ है, तो 'k' का मान ज्ञात कीजिए।

Ans. 25

Sol. $2x = y^{\frac{1}{5}} + y^{-\frac{1}{5}}$

Differentiating with respect to y

$$\Rightarrow 2\frac{dx}{dy} = \frac{1}{5}y^{\frac{1}{5}-1} - \frac{1}{5}y^{-\frac{1}{5}-1}$$

$$= \frac{1}{5y}\left(y^{\frac{1}{5}} - y^{-\frac{1}{5}}\right)$$

$$= \frac{1}{5y}\sqrt{\left(y^{\frac{1}{5}} + y^{-\frac{1}{5}}\right)^2 - 4}$$

$$\Rightarrow 10y\frac{dx}{dy} = \sqrt{4x^2-4}$$

$$\Rightarrow 5y = \sqrt{x^2-1}\frac{dy}{dx} \qquad \text{_____(1)}$$

differentiating with respect to x

$$5\frac{dy}{dx} = \sqrt{x^2-1}\frac{d^2y}{dx^2} + \frac{x}{\sqrt{x^2-1}}\frac{dy}{dx}$$

$$\left(x^2-1\right)\frac{d^2y}{dx^2}+x\frac{dy}{dx}=5\sqrt{x^2-1}\frac{dy}{dx}=25y$$

$k = 25$

4. The function $f: R \to R$ satisfies $f(x^2) \cdot f''(x) = f'(x) \cdot f'(x^2)$ for all real x. Given that $f(1) = 1$ and $f'''(1) = 8$, compute the value of $f'(1) + f''(1)$.

फलन $f: R \to R$, सम्बन्ध $f(x^2) \cdot f''(x) = f'(x) \cdot f'(x^2)$ को सभी वास्तविक x के लिये संतुष्ट करता है। दिया गया है कि $f(1) = 1$ तथा $f'''(1) = 8$ है, तो $f'(1) + f''(1)$ का मान ज्ञात कीजिए।

Ans. 6

Sol. $f\left(x^2\right)\cdot f''(x)=f'(x)\cdot f'\left(x^2\right)$ _____(1)

put $x = 1$ $f(1) = 1$ $f'''(1) = 8$

$f(1)\, f''(1)=f'(1)^2$

$f'(1)^2 = f''(1)$ _____(2)

Differentiating with respect to x of (1)

$f(x^2)\, f'''(x)+f'(x^2)\cdot 2x\cdot f''(x)=f''(x)\cdot f'(x^2)+f'(x)\cdot 2x\cdot f''(x^2)$

Put $x = 1$

$f(1)\cdot f'''(1)+f'(1)\, 2\cdot f''(1)=f''(1)\cdot f'(1)+f'(1)\cdot 2\cdot f''(1)$

$8 = f''(1)\cdot f'(1)$ _____(3)

from (1) and (3) $f'(1) = 2$

$f''(1) = 4$

$f'(1) + f''(1) = 6$

5. (a) Show that the substitution $z = \ell n\left(\tan\frac{x}{2}\right)$ changes the equation $\frac{d^2y}{dx^2}+\cot x\frac{dy}{dx}+4y\cos ec^2x=0$ to $(d^2y/dz^2) + 4y = 0$.

सिद्ध कीजिये कि प्रतिस्थापन $z = \ell n\left(\tan\frac{x}{2}\right)$ समीकरण $\frac{d^2y}{dx^2}+\cot x\frac{dy}{dx}+4y\cos ec^2x=0$ को $(d^2y/dz^2) + 4y = 0$ में परिवर्तित कर देगा।

Ans.

Sol. $z = \ell n\left(\tan\frac{x}{2}\right)$

$$\frac{dz}{dx} = \frac{1}{\sin x} \qquad \text{Now } \frac{dy}{dz} = \frac{dy}{dx}.\frac{dx}{dz} = \frac{dy}{dx}\sin x$$

$$\frac{d^2y}{dz^2} = \frac{d}{dx}\left(\frac{dy}{dx}\sin x\right).\frac{dx}{dz}$$

$$\frac{d^2y}{dz^2} = \left(\sin x\frac{d^2y}{dx^2} + \cos x\frac{dy}{dx}\right)\sin x \quad .....(i)$$

$$\frac{d^2y}{dz^2} = \sin^2 x\frac{d^2y}{dx^2} + \sin x\cos x\frac{dy}{dx}$$

$$\text{cosec}^2 x\frac{d^2y}{dz^2} = \frac{d^2y}{dx^2} + \cot x\frac{dy}{dx}$$

Now put value from given equation

$$\text{cosec}^2x\frac{d^2y}{dz^2} + 4y\text{cosec}^2x = 0$$

$$\frac{d^2y}{dz^2} + 4y = 0$$

(b) If the dependent variable y is changed to 'z' by the substitution $y = \tan z$ then the differential equation

$\frac{d^2y}{dx^2} = 1 + \frac{2(1+y)}{1+y^2}\left(\frac{dy}{dx}\right)^2$ is changed to $\frac{d^2z}{dx^2} = \cos^2 z + k\left(\frac{dz}{dx}\right)^2$, then find the value of *k*.

यदि परतंत्र चर y को प्रतिस्थापन $y = \tan z$ द्वारा 'z' में परिवर्तित कर दिया जाए तो अवकल समीकरण $\frac{d^2y}{dx^2} = 1 + \frac{2(1+y)}{1+y^2}\left(\frac{dy}{dx}\right)^2$,

$\frac{d^2z}{dx^2} = \cos^2 z + k\left(\frac{dz}{dx}\right)^2$ में बदल जाएगी, तो k का मान बताइए।

Ans. $k = 2$

Sol. $y = \tan z$ _____(1)

$$\frac{dy}{dx} = \sec^2 z.\frac{dz}{dx} \qquad \text{_____(2)}$$

Diff^n w.r.t. 'x'

$$\frac{d^2y}{dx^2} = \sec^2 z \cdot \frac{d^2z}{dx^2} + 2\sec^2 z \cdot \tan z \left(\frac{dz}{dx}\right)^2 \qquad \text{_____(3)}$$

put values from (1), (2) and (3)

in given eqn $\quad \frac{d^2y}{dx^2} = 1 + \frac{2(1+y)}{1+y^2}\left(\frac{dy}{dx}\right)^2$

$$\Rightarrow \sec^2 z \cdot \frac{d^2z}{dx^2} + 2\sec^2 z \cdot \tan z \left(\frac{dz}{dx}\right)^2$$

$$= 1 + \frac{2(1+\tan z)}{\sec^2 z}\left(\sec^2 z \frac{dz}{dx}\right)^2$$

$$\Rightarrow \frac{d^2z}{dx^2} + 2\tan z \left(\frac{dz}{dx}\right)^2 = \cos^2 z + 2(1+\tan z)\left(\frac{dz}{dx}\right)^2$$

$$\Rightarrow \frac{d^2z}{dx^2} = \cos^2 z + 2\left(\frac{dz}{dx}\right)^2$$

Hence k = 2

6. Show that $R = \frac{\left[1+(dy/dx)^2\right]^{\frac{3}{2}}}{d^2y/dx^2}$ can be reduced to the form $R^{2/3} = \frac{1}{\left(d^2y/dx^2\right)^{2/3}} + \frac{1}{\left(d^2x/dy^2\right)^{2/3}}$.

सिद्ध कीजिये की $R = \frac{\left[1+(dy/dx)^2\right]^{\frac{3}{2}}}{d^2y/dx^2}$ को $R^{2/3} = \frac{1}{\left(d^2y/dx^2\right)^{2/3}} + \frac{1}{\left(d^2x/dy^2\right)^{2/3}}$ में विघटित किया जा सकता है।

Ans.

Sol. $$R = \frac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}}}{\left(\frac{d^2y}{dx^2}\right)} \qquad \text{_____(1)}$$

Let $\frac{dy}{dx} = \tan\theta \qquad \Rightarrow \frac{d^2y}{dx^2} = \sec^2\theta.\frac{d\theta}{dx}$

$$\Rightarrow \frac{1}{\frac{d^2y}{dx^2}} = \cos^2\theta.\frac{dx}{d\theta}$$

Now from $R = \sec^3\theta\cos^2\theta\cdot\frac{dx}{d\theta}$

$R = \sec\theta\frac{dx}{d\theta}$ $\Rightarrow \frac{d\theta}{dx} = \frac{\sec\theta}{R}$

$\frac{d^2y}{dx^2} = \frac{\sec^3\theta}{R}$ _____(2)

$\frac{dx}{dy} = \cot\theta \Rightarrow \frac{d^2x}{dy^2} = -\cos ec^2\theta\cdot\frac{d\theta}{dy}$

$= -\cos\alpha^2\theta\cdot\frac{d\theta}{dx}\cdot\frac{dx}{dy}$

$\frac{d^2x}{dy^2} = -\cos ec^2\theta\cdot\frac{\sec\theta}{R}\cdot\cot\theta = -\frac{\cos ec^3\theta}{R}$ _____(3)

Now $\frac{1}{\left(\frac{d^2y}{dx^2}\right)^{\frac{2}{3}}} + \frac{1}{\left(\frac{d^2x}{dy^2}\right)^{\frac{2}{3}}} = R^{\frac{2}{3}}\left[\sin^2\theta + \cos^2\theta\right]$

$= R^{\frac{2}{3}}$

7. Suppose $p(x) = a_0 + a_1 x + a_2 x^2 + \ldots\ldots + a_n x^n$. If $|p(x)| \le |e^{x-1} - 1|$ for all $x \ge 0$ prove that $|a_1 + 2a_2 + \ldots\ldots + n a_n| \le 1$. **[JEE-2000]**

माना $p(x) = a_0 + a_1 x + a_2 x^2 + \ldots\ldots + a_n x^n$ है। यदि $x \ge 0$ के लिए $|p(x)| \le |e^{x-1} - 1|$ हो, तो सिद्ध कीजिए कि $|a_1 + 2a_2 + \ldots\ldots + n a_n| \le 1$. **[JEE-2000]**

Ans.

Sol. $p(x) = a_0 + a_1x + a_2x^2 + \ldots\ldots + a_nx^n$

$\Rightarrow \quad p'(x) = a_1 + 2a_2x + \ldots\ldots + na_nx^{n-1}$

$\Rightarrow \quad p'(1) = a_1 + 2a_2 + \ldots\ldots + na_n \quad \ldots\ldots(i)$

Now $|p(x)| \le |e^{x-1} - 1| \ \forall\ x \ge 0$ (given)

$\Rightarrow \quad |p(1)| \le |e^0 - 1| = |1 - 1| = 0$

But $\quad |p(1)| \ge 0 \quad \Rightarrow \quad p(1) = 0$

Now for $\quad -1 < h < \infty$, $h \ne 0$, $1 + h > 0$

and $\quad |p(x)| \le |e^{x-1} - 1| \ \forall \ x \ge 0$

$\Rightarrow \quad |p(1 + h)| \le |e^h - 1| \ \forall \ h > -1, h \ne 0$

$\Rightarrow \quad |p(1 + h) - p(1)| \le |e^h - 1| \quad \because \quad p(1) = 0$

$$\Rightarrow \quad \left|\frac{p(1+h)-p(1)}{h}\right| \le \left|\frac{e^h - 1}{h}\right|$$

Taking limit as $h \to 0$, we get

$$\lim_{h\to 0}\left|\frac{p(1+h)-p(1)}{h}\right| \le \lim_{h\to 0}\left|\frac{e^h - 1}{h}\right| \quad \Rightarrow \quad |p'(1)| \le 1$$

$\Rightarrow \quad |a_1 + 2a_2 + 3a_3 + \ldots\ldots\ldots na_n| \le 1 \qquad \{\text{from (i)}\}$

Hence proved

8. Suppose f and g are two functions such that $f, g : R \to R, f(x) = \ell n\left(1 + \sqrt{1 + x^2}\right)$ and $g(x) = \ell n\left(x + \sqrt{1 + x^2}\right)$

then find the value of $x\, e^{g(x)}\left(f\left(\frac{1}{x}\right)\right)' + g'(x)$ at $x = 1$.

माना f तथा g दो फलन इस प्रकार है कि $f, g : R \to R$, $f(x) = \ell n\left(1 + \sqrt{1 + x^2}\right)$ तथा $g(x) = \ell n\left(x + \sqrt{1 + x^2}\right)$ तो

$x\, e^{g(x)}\left(f\left(\frac{1}{x}\right)\right)' + g'(x)$ का $x = 1$ पर मान निकालिए।

Ans. Zero

Sol. $$\left[f\left(\frac{1}{x}\right)\right]' = f'\left(\frac{1}{x}\right)\cdot\left(-\frac{1}{x^2}\right) \qquad \text{_____(1)}$$

$$f(x) = \ell n\left(1 + \sqrt{1 + x^2}\right)$$

$$f'(x) = \frac{1}{1 + \sqrt{1 + x^2}} \cdot \frac{x}{\sqrt{1 + x^2}}$$

$$f'\left(\frac{1}{x}\right)=\frac{x}{x+\sqrt{1+x^2}}\cdot\frac{1}{\sqrt{1+x^2}}$$

$$e^{g(x)}f'\left(\frac{1}{x}\right)=\frac{x}{\sqrt{1+x^2}}$$

$$x.e^{g(x)}\cdot f'\left(\frac{1}{x}\right)=\frac{x^2}{\sqrt{1+x^2}}$$

$$g(x)=\ell n\left(x+\sqrt{1+x^2}\right)$$

$$g'(x)=\frac{1}{\sqrt{1+x^2}}$$

Now $x.e^{g(x)}\left[f\left(\frac{1}{x}\right)\right]^1+g'(x)$

$$=x.e^{g(x)}\cdot f'\left(\frac{1}{x}\right)\cdot\left(-\frac{1}{x^2}\right)+g'(x)$$

$$=-\frac{1}{\sqrt{1+x^2}}+\frac{1}{\sqrt{1+x^2}}=0$$

$$\Rightarrow x.e^{g(x)}\left[f\left(\frac{1}{k}\right)\right]^1+g'(x)=0$$

then $\left[x.e^{g(x)}\left[f\left(\frac{1}{x}\right)\right]^1+g'(x)\right]_{x=1}=0$

9. Let $f(x)=\begin{bmatrix} xe^x & x\le 0 \\ x+x^2-x^3 & x>0 \end{bmatrix}$ then prove that :

(a) f is continuous and differentiable for all x.

(b) f' is continuous and differentiable for all x.

माना $f(x)=\begin{bmatrix} xe^x & x\le 0 \\ x+x^2-x^3 & x>0 \end{bmatrix}$ तो सिद्ध कीजिये। :

(a) f सभी x के मानों के लिये सतत् एवं अवकलनीय है।

(b) f' सभी x के मानों के लिये सतत् एवं अवकलनीय है।

Ans.

Sol. $f(x)=\begin{cases} x\cdot e^{x} & x\le 0 \\ x+x^{2}-x^{3} & x\ge 0 \end{cases}$

$\because x.e^{x}$ cont^{n} and diff^{n} $\forall x\in R$

$\therefore xe^{x}$ cont^{n} and diff^{n} $\forall x>0$

it means

$f(x)=\begin{cases} x.e^{x} & x\le 0 \\ x+x^{2}-x^{3} & x>0 \end{cases}$ contn and diff $\forall x\in R-\{0\}$

Now we check cont^{n} and diff^{n} at $x=0$

$\left.\begin{array}{l} f(0-0)=0 \\ f(0)=0 \\ f(0+0)=0 \end{array}\right\}$ f is cont^{n} at $x=0$

$f'(0-0)=1+0=1$ = L.H.D

$f'(0+0)=1+2(0)-3(0)=1=$ R.H.D

f is diff at $x=0$

Hence t is cont^{n} and diff^{n} at $x=0$

$g(x)=f'(x)\begin{cases} xe^{x}+e^{x} & x\le 0 \\ 1+2x-3x^{2} & x>0 \end{cases}$

$f'(x)$ cont^{n} and diff $\forall x\in R-\{0\}$

Now we check cont^{n} and diff^{n} at $x=0$

$\left.\begin{array}{l} g(0-0)=1=\text{L.H.L} \\ g(0)=1 \\ g(0+0)=1=\text{R.H.L} \end{array}\right\}$ $g=f'$ cont^{n} at $x=0$

hence f' cont^{n} $\forall x\in R$

$\left.\begin{array}{l} g'(0-0)=2=\text{L.H.D} \\ g'(0+0)=2=\text{R.H.D} \end{array}\right\}$ $g=f'$ diff^{n} at $x=0$

Hence f ' diff $\forall x \in R$

10. Let $f(x)=\begin{vmatrix} a+x & b+x & c+x \\ \ell+x & m+x & n+x \\ p+x & q+x & r+x \end{vmatrix}$. Show that $f''(x)=0$ and that $f(x)=f(0)+kx$ where k denotes the sum of all the co-factors of the elements in $f(0)$.

माना $f(x)=\begin{vmatrix} a+x & b+x & c+x \\ \ell+x & m+x & n+x \\ p+x & q+x & r+x \end{vmatrix}$ तो सिद्ध कीजिये की $f''(x)=0$ तथा $f(x)=f(0)+kx$ जहाँ k, सारणिक $f(0)$ के सभी अवयवों के सह खण्डजो का योग है।

Ans.

Sol. $f(x)=\begin{vmatrix} a+x & b+x & c+x \\ \ell+x & m+x & n+x \\ p+x & q+x & r+x \end{vmatrix}$

$c_3 \to c_3 - c_1 \quad c_2 \to c_2 - c_1$

$$f(x)=\begin{vmatrix} a+x & b-a & c-a \\ \ell+x & m-\ell & n-\ell \\ p+x & q-p & r-p \end{vmatrix}$$

$$f'(x)=\begin{vmatrix} 1 & b-a & c-a \\ 1 & m-\ell & n-\ell \\ 1 & q-p & r-p \end{vmatrix}+0+0=k$$

$\Rightarrow f'(x)=k$ _____(1)

Differentiate w.r.t. x

$\Rightarrow f''(x)=0$ Hence proved

$$\Rightarrow k=\begin{vmatrix} 1 & b+x & c+x \\ 1 & m+x & n+x \\ 1 & q+x & r+x \end{vmatrix}+\begin{vmatrix} a+x & 1 & c+x \\ \ell+x & 1 & n+x \\ p+x & 1 & r+x \end{vmatrix}+\begin{vmatrix} a+x & b+x & 1 \\ \ell+x & m+x & 1 \\ p+x & q+x & 1 \end{vmatrix}$$

$c_2 \to c_2 - xc_1$ $\quad$ $c_3 + c_3 - x_4$

Now $\begin{vmatrix} 1 & b+x & c+x \\ 1 & m+x & n+x \\ 1 & q-x & r+x \end{vmatrix} = \begin{vmatrix} 1 & b & c \\ 1 & m & n \\ 1 & q & r \end{vmatrix}$ $\quad$ $\left\{\begin{matrix} \text{apply} & c_2 \to c_2 - xc_1 \\ & c_3 \to c_3 - xc_1 \end{matrix}\right\}$

$= (mr - qn) - (br - qc) + (bn + cm)$

$=$ sum of cofactor of a_{11}, a_{21}, a_{31} in f(0)

Illy

$$\begin{vmatrix} a+x & 1 & c+x \\ \ell+x & 1 & n+x \\ p+x & 1 & r+x \end{vmatrix} = \text{sum of cofactor of } a_{12}, a_{22}, a_{32} \text{ of f(0)}$$

And

$$\begin{vmatrix} a+x & b+x & 1 \\ \ell+x & m+x & 1 \\ p+x & q+x & 1 \end{vmatrix} = \text{sum of cofactor of } a_{13}, a_{23}, a_{33} \text{ of f(0)}$$

Hence $\quad k =$ sum of cofactor of all elements of s(0)

$\Rightarrow$ from eq. (1) $\quad f'(x) = k$

Integrate

$\Rightarrow f(x) = kx + c$ $\quad$ put $x = 0$ $\quad$ $c = f(0)$

then $f(x) = kx + f(0)$ $\quad$ H.P.

11. Let $a_1 > a_2 > a_3 \ldots\ldots\ldots a_n > 1$; $p_1 > p_2 > p_3 \ldots\ldots\ldots > p_n > 0$; such that $p_1 + p_2 + p_3 + \ldots\ldots + p_n = 1$

Also $F(x) = \left(p_1 a_1^x + p_2 a_2^x + \ldots\ldots + p_n a_n^x\right)^{\frac{1}{x}}$. Compute

माना $a_1 > a_2 > a_3 \ldots\ldots\ldots a_n > 1$; $p_1 > p_2 > p_3 \ldots\ldots\ldots > p_n > 0$; जैसे कि $p_1 + p_2 + p_3 + \ldots\ldots + p_n = 1$ है, साथ ही $F(x) = \left(p_1 a_1^x + p_2 a_2^x + \ldots\ldots + p_n a_n^x\right)^{1/x}$ है तो ज्ञात कीजिये।

(a) $\lim\limits_{x \to 0^+} F(x)$ $\quad$ (b) $\lim\limits_{x \to \infty} F(x)$ $\quad$ (c) $\lim\limits_{x \to -\infty} F(x)$

Ans. (a) $a_1^{p_1} \cdot a_2^{p_2} .....a_n^{p_n}$; (b) a_1 ; (c) a_n

Sol. (a) $F(x) = \left(p_1 a_1^x + p_2 a_2^x + ... + p_n a_n^x\right)^{\frac{1}{x}}$

$$\lim_{x \to 0^+} F(x) = \lim_{x \to 0^+} \left(p_1 a_1^x + p_2 a_2^x + ... + p_n a_n^x\right)^{\frac{1}{x}} \qquad \left[1^\infty\right]$$

$$\lim_{x \to 0^+} F(x) = e^{\lim_{x \to 0} \frac{\left(p_1 a_1^x + p_2 a_2^x + ... + p_n a_n^x - 1\right)}{x}} \qquad [\%]$$

$$= e^{\lim_{x \to 0} p_1 a_1^x . \ell n a_1 + p_2 a_a^x \ell n a_2 + ... + p_n a_n^x . \ell n a_2} \qquad [\text{By L} - \text{hospital rule}]$$

$$= e^{\ell n\left(a_1^{p_1} \cdot a_2^{p_2} .....a_n^{p_n}\right)} = a_1^{p_1} \cdot a_2^{p_2} .....a_n^{p_n}$$

(b) $a_1 > a_2 > a_3 > .....a_n > 1$

$$\Rightarrow \frac{a_2}{a_1} < 1, \frac{a_3}{a_1} < 1, ..., \frac{a_n}{a_1} < 1$$

$$\Rightarrow \left(\frac{a_i}{a_1}\right)^\infty = 0 \quad \forall i \in \{2, 3, ...n\}$$

$$\lim_{x \to \infty} F(x) = \lim_{x \to \infty} a_1 \left[p_1 + p_2 \left(\frac{a_2}{a_1}\right)^x + ... + p_n \left(\frac{a_n}{a_1}\right)^x\right]^{\frac{1}{x}}$$

$$= a_1 \left[p_1 + 0 + 0 + ... + 0\right]^0 = a_1$$

(c) $\frac{a_1}{a_n} > 1, \frac{a_2}{a_n} > 1, \frac{a_3}{a_n} > 1, ... \frac{a_{n-1}}{a_n} > 1$

$$\left(\frac{a_i}{a_n}\right)^{-\infty} = 0 \qquad \forall i \in \{1, 2, 3, ..., n-1\}$$

$$\lim_{x \to -\infty} F(x) = \lim_{x \to -\infty} a_n \left[p_1 \left(\frac{a_1}{a_n}\right)^x + p_2 \left(\frac{a_2}{a_n}\right)^n + ... + p_{n-1} \left(\frac{a_n - 1}{a_n}\right)^n + p_n\right]^{\frac{1}{x}}$$

$$= a_n \left[0 + 0 + ... + 0 + p_n\right]^\circ$$

$= a_n$

12. If $x_1, x_1, x_2, x_3, \ldots\ldots x_{n-1}$ be n zero's of the polynomial $P(x) = x^n + \alpha x + \beta$, where $x_i \neq x_j \ \forall$ i and $j = 1, 2, 3, \ldots (n-1)$. Prove that the value of $Q(x) = (x_1 - x_2)(x_1 - x_3)(x_1 - x_4) \ldots\ldots (x_1 - x_{n-1})$, is equal to ${}^nC_2\ x_1^{n-2}$.

यदि $x_1, x_1, x_2, x_3, \ldots\ldots x_{n-1}$ बहुपद $P(x) = x^n + \alpha x + \beta$ के 'n' मूल है जहाँ $x_i \neq x_j \ \forall$ i तथा $j = 1, 2, 3, \ldots (n-1)$ तो सिद्ध कीजिये $Q(x) = (x_1 - x_2)(x_1 - x_3)(x_1 - x_4) \ldots\ldots (x_1 - x_{n-1})$ का मान ${}^nC_2\ x_1^{n-2}$ के बराबर है।

Ans.

Sol. $P(x) = x^n + \alpha x + \beta = (x - x_1)^2 (x - x_3) \ldots\ldots (x - x_{n-1})$

$$\Rightarrow \frac{x^n + \alpha x + \beta}{(x - x_1)^2} = (x - x_2)(x - x_3) \ldots\ldots (x - x_{n-1})$$

$$\Rightarrow \lim_{x \to x_1} \frac{x^n + \alpha x + \beta}{(x - x_n)^2} = (x_1 - x_2)(x_1 - x_3) \ldots\ldots (x_1 - x_{n-1})$$

$$\Rightarrow \lim_{x \to x_1} \frac{nx^{n-1} + \alpha}{2(x - x_1)} = Q(x) \qquad \text{[By L – Hospital rule]}$$

$$\Rightarrow \frac{n(n-1)}{2} x_1^{n-2} = Q(x) \qquad \text{[Again by L – Hospital rule]}$$

$$\Rightarrow {}^nC_2 \cdot x_1^{n-2} = Q(x) \quad \text{H.P.}$$

13. If $\frac{d^2}{dx^2}\left(\frac{\sin^4 x + \sin^2 x + 1}{\sin^2 x + \sin x + 1}\right) = a\sin^2 x + b\sin x + c$ then the value of $b + c - a$ is.

यदि $\frac{d^2}{dx^2}\left(\frac{\sin^4 x + \sin^2 x + 1}{\sin^2 x + \sin x + 1}\right) = a\sin^2 x + b\sin x + c$ है तो $b + c - a$ का मान होगा?

Ans. 7

Sol. $P^4 + P^2 + 1 = (P^2 - P + 1)(P^2 + P + 1)$

$$\Rightarrow \frac{P^4 + P^2 + 1}{P^2 + P + 1} = P^2 - P + 1$$

$$\frac{\sin^4 x+\sin^2 x+1}{\sin^2 x+\sin x+1}=\sin^2-\sin x+1$$

Now $\frac{d^2}{dx^2}\left[\frac{\sin^4 x+\sin^2 x+1}{\sin^2 x+\sin x+1}\right]=2\cos 2x+\sin x$

$$=2\left(1-2\sin^2 x\right)+\sin x$$

$$=-4\sin^2 x+\sin x+2$$

Compairing with $a\sin^2x + b\sin x + c$

$a = -4 \qquad b = 1 \quad c = 2$

$b + c - a = 7$

14. Let $f(x)=x+\frac{x^2}{2}+\frac{x^3}{3}+\frac{x^4}{4}+\frac{x^5}{5}$ and let $g(x) = f^{-1}(x)$. Find $g'''(0)$.

माना $f(x)=x+\frac{x^2}{2}+\frac{x^3}{3}+\frac{x^4}{4}+\frac{x^5}{5}$ तथा $g(x) = f^{-1}(x)$ है तो $g'''(0)$ का मान ज्ञात कीजिये ?

Ans. 1

Sol. $f(x)=x+\frac{x^2}{2}+\frac{x^3}{3}+\frac{x^4}{4}+\frac{x^5}{5} \qquad \Rightarrow f(0)=0$

$$f'(x)=1+x+x^2+x^3+x^4=\frac{x^5-1}{x-1}$$

$$f''(x)=1+2x+3x^2+4x^3$$

$$f'''(x)=2+6x+12x^2$$

$$f'(0)=1, \qquad f''(0)=1, \qquad f'''(0)=2$$

$g(x)=f^{-1}(x) \Rightarrow f[g(x)]=x$ and $g[f(x)]=x$

$\Rightarrow g[f(x)]=x$

Diff^n w.r.t. x

$\Rightarrow g'\left[f(x)\right]\ f'(x)=1$ _____(1) $\rightarrow g'(0)=1$

Again Diff^n w.r.t. x

$\Rightarrow g''\left(f(x)\right)\cdot\left\{f'(x)\right\}^2+g'\left(f(x)\right)\cdot f''(x)=0$ _____(2)

put $x=0$

$\Rightarrow g''\left(f(0)\right)\cdot\left\{f'(0)\right\}^2+g'\left(f(0)\right)\cdot f''(0)=0$

$\Rightarrow g''(0)+g'(0)=0$

$g''(0)=-1$

Again diff^n w.r.t. x of eq^n (1)

$\Rightarrow g''\left(f(x)\right)\cdot\left\{f'(x)\right\}^3+g''\left(f(x)\right)\cdot 2f'(x)\cdot f''(x)+g''\left(f(x)\right)\cdot f'(x)\cdot f''(x)+g'\left(f(x)\right)\cdot f''(x)=0$

put $x=0$

$\Rightarrow g'''(0)+g''(0)2+g''(0)+g'(0)\cdot(2)=0$

$g'''(0)+3(-1)+2=0$

$g'''(0)=1$

15. The value of x, $x\in(2,\infty)$ where $f(x)=\sqrt{x+\sqrt{8x-16}}+\sqrt{x-\sqrt{8x-16}}$ is not differentiable is.

ग $\in(2,\infty)$ में x का मान ज्ञात कीजिये जहां $f(x)=\sqrt{x+\sqrt{8x-16}}+\sqrt{x-\sqrt{8x-16}}$ अवकलनीय नहीं है ?

Ans. 4

Sol. $f(x)=\sqrt{x+\sqrt{8x-16}}+\sqrt{x-\sqrt{8x-16}}$

Let $8x-16=p^2$ $\Rightarrow x=\frac{p^2+16}{8},$ $p>0$

$f(x)=\frac{\sqrt{p^2+16+8p}+\sqrt{p^2+16-8p}}{2\sqrt{2}}$

$$= \frac{|p+4|+|p-4|}{2\sqrt{2}} = \frac{p+4+|p-4|}{2\sqrt{2}}$$

$$f(x) = \begin{cases} \frac{2p}{2\sqrt{2}} & p \geq 4 \quad \Rightarrow 8x \geq 32 \quad \Rightarrow x \geq 4 \\ 2\sqrt{2} & P < 4 \quad \Rightarrow 2 < X < 4 \end{cases}$$

$$f'(4+0) = \frac{1}{2\sqrt{16}} \cdot \frac{8}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$f'(4-0) = 0$$

f is not differential at x = 4

16. If $y = \left[\ell n \left(\frac{x+\sqrt{x^2-a^2}}{a} \right) \right]^2 + k\, \ell n \left(x+\sqrt{x^2-a^2} \right)$, then find the value of $(x^2-a^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx}$:

यदि $y = \left[\ell n \left(\frac{x+\sqrt{x^2-a^2}}{a} \right) \right]^2 + k\, \ell n \left(x+\sqrt{x^2-a^2} \right)$ तो $(x^2-a^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx}$ का मान ज्ञात कीजिए :

Ans. 2

Sol. $$y = \left[\ell n \left(\frac{x+\sqrt{x^2-a^2}}{a} \right) \right]^2 + k\, \ell n \left(x+\sqrt{x^2-a^2} \right)$$

Differentiating both sides w. r. t.

$$\frac{dy}{dx} = 2\ell n \left(\frac{x+\sqrt{x^2-a^2}}{a} \right) \frac{1}{\sqrt{x^2-a^2}} + \frac{k}{\sqrt{x^2-a^2}}$$

$$\sqrt{x^2-a^2} \cdot \frac{dy}{dx} = 2\ell n \left(\frac{x+\sqrt{x^2-a^2}}{a} \right) + k$$

Differentiating w.r.t. x

$$\sqrt{x^2-a^2}\,\frac{d^2y}{dx^2} + \frac{x}{\sqrt{x^2-a^2}} \cdot \frac{dy}{dx} = \frac{2}{\sqrt{x^2-a^2}}$$

$$\Rightarrow \left(x^2 - a^2\right)\frac{d^2y}{dx^2} + x\frac{dy}{dx} = 2$$

17. If $(a + bx)\, e^{y/x} = x$, show that $x^3 y'' = (xy' - y)^2$ _____.

यदि $(a + bx)\, e^{y/x} = x$ तो प्रदर्शित कीजिए कि $x^3 y'' = (xy' - y)^2$ _____.

Ans.

Sol. $e^{\frac{y}{x}} = \dfrac{x}{a+bx}$ $\Rightarrow$ taking log on both side

$$\Rightarrow \frac{y}{x} = \ell n\left(\frac{x}{a+bx}\right)$$

$$\Rightarrow y = x\cdot\ell n\left(\frac{x}{a+bx}\right)$$

Differentiating w.r.t. "x"

$$\Rightarrow \frac{dy}{dx} = \ell n\left(\frac{x}{a+bx}\right) + x\left[\frac{1}{x} - \frac{b}{a+bx}\right]$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{x} + \frac{x[a]}{x(a+bx)} \qquad \text{_____(1)}$$

$$\Rightarrow x\cdot\frac{dy}{dx} - y = \frac{ax}{a+bx} = ae^{\frac{y}{x}} \qquad \text{_____(2)}$$

from (1)

$$\frac{d^2y}{dx^2} = \frac{xy' - y}{x^2} - \frac{ab}{(a+bx)^2}$$

$$\Rightarrow x^3\frac{d^2y}{dx^2} = (xy' - y)\cdot x - \frac{b}{a}x\left(\frac{ax}{a+bx}\right)^2$$

$$\Rightarrow (xy' - y)x - \frac{b}{a}x(xy' - y)^2$$

$$\Rightarrow x(xy' - y)\left[1 - \frac{b}{a}\frac{ax}{a+bx}\right]$$

$$\Rightarrow (xy'-y)\frac{ax}{a+bx}$$

$$\Rightarrow x^3y''' = (xy'-y)^2$$

18. If $\frac{x+b}{2} = a\tan^{-1}(a\,\ell n\,y)$, $a>0$ prove that $yy'' - yy'\ \ell n\,y = (y')^2$ _____.

यदि $\frac{x+b}{2} = a\tan^{-1}(a\,\ell n\,y)$, $a>0$ तो सिद्ध करो कि $y\,y'' - yy'\ \ell n\,y = (y')^2$ _____.

Ans.

Sol. $\frac{x+b}{2} = a\tan^{-1}(a\ell ny) \qquad a>0$

$$\Rightarrow \tan\left(\frac{x+b}{2a}\right) - a\ell ny = 0$$

$$\because \frac{dy}{dx} = -\frac{\delta f/dx}{\delta f/dy} = -\frac{\frac{1}{2a}\cdot\sec^2\left(\frac{x+b}{2a}\right)}{-\frac{a}{y}}$$

$$\Rightarrow \frac{2a^2y'}{y} = 1+\tan^2\left(\frac{x+b}{2a}\right)$$

$$\Rightarrow \frac{2a^2\cdot y'}{y} = 1+a^2\ell n^2 y$$

Differentiating w.r.t. x

$$\Rightarrow 2a^2\cdot\frac{yy''-(y')^2}{y^2} = 2a^2\cdot\ell ny\cdot\frac{1}{y}\cdot y'$$

$$\Rightarrow yy''-(y')^2 = y\cdot y'\ell ny$$

$$\Rightarrow yy''-yy'\ell ny = (y')^2$$

19. If $y^3 - y = 2x$, then prove that $\frac{d^2y}{dx^2} = -\frac{24y}{(3y^2-1)^3}$. Hence, show that $\left(x^2 - \frac{1}{27}\right)\frac{d^2y}{dx^2} + x\frac{dy}{dx} = \frac{y}{9}$ _____.

यदि $y^3 - y = 2x$ तो सिद्ध करो कि $\frac{d^2y}{dx^2} = -\frac{24y}{(3y^2-1)^3}$ तथा इस प्रकार प्रदर्शित कीजिए कि $\left(x^2 - \frac{1}{27}\right)\frac{d^2y}{dx^2} + x\frac{dy}{dx} = \frac{y}{9}$ _____.

Ans.

Sol. $y^3 - y = 2x \qquad y^2 - 1 = \frac{2x}{y}$

$$\Rightarrow \left(3y^2 - 1\right)\frac{dy}{dx} = 2$$

$$\Rightarrow \frac{dy}{dx} = \frac{2}{3y^2 - 1} = \frac{2}{\alpha}(\text{say})$$

$$\Rightarrow \frac{d^2y}{dx^2} = -\frac{12y\cdot\frac{dy}{dx}}{\left(3y^2-1\right)} = -\frac{12y}{\left(3y^2-1\right)^2}\cdot\frac{2}{\left(3y^2-1\right)}$$

$$\Rightarrow \frac{d^2y}{dx^2} = -\frac{24y}{\left(3y^2-1\right)^3} \qquad 3y^2 - \alpha \qquad y^2 = \frac{\alpha+1}{3} \qquad y^2 - 1 = \frac{\alpha-2}{3}$$

Now $\text{L.H.S.} = \left(x^2 - \frac{1}{27}\right)\frac{d^2y}{dx^2} + x\frac{dy}{dx}$

$$= \left[\frac{y^2\left(y^2-1\right)^2}{4} - \frac{1}{27}\right]\left[\frac{-24y}{\left(3y^2-1\right)^3}\right] \times \frac{y\left(y^2-1\right)}{2}\cdot\frac{2}{3y^2-1}$$

$$= y\left[\left[\frac{93y^2\left(y^2-1\right)^2 - 4}{108}\right]\left[\frac{-24}{\left(3y^2-1\right)^3}\right] + \frac{y^2-1}{3y^2-1}\right]$$

$$= y\left[\frac{\left(\frac{9(\alpha+1)(\alpha-2)^2}{9}-4\right)}{108}\left(-\frac{24}{\alpha^3}\right)+\frac{\alpha-2}{3\alpha}\right]$$

$$= \frac{y}{9}\left[\left((\alpha+1)(\alpha-2)^2-4\right)\left(-\frac{2}{\alpha^3}\right)+\frac{3\alpha-6}{\alpha}\right]$$

$$= \frac{y}{9}\left[\frac{8-2(\alpha+1)(\alpha^2-4\alpha+4)+3\alpha^3-6\alpha^2}{\alpha^3}\right]$$

$$= \frac{y}{9}\left[\frac{8-2\alpha^3+6\alpha^2-8+3\alpha^3-6\alpha^2}{\alpha^3}\right]$$

$$= \frac{y}{9} = \text{R.H.S.}$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 20 to 22

In certain problem the differentiation of $\{f(x) \cdot g(x)\}$ appears. One student commits mistake and differentiates as $\frac{df}{dx}\cdot\frac{dg}{dx}$ but he gets correct result if $f(x) = x^3$ and $g(x)$ is a decreasing function for which $g(0) = \frac{1}{3}$.

किसी प्रश्न में $\{f(x) \cdot g(x)\}$ का अवकलन पूछा जाता है। एक विद्यार्थी गलती करके इसका अवकलन $\frac{df}{dx}\cdot\frac{dg}{dx}$ कर देता है परन्तु उसे

l ghi fj . kke i kir gksrk gS] fn $f(x) = x^3$ तथा $g(x)$ एक ह्रासमान फलन है जिसके लिये $g(0) = \frac{1}{3}$

20. The function $g(x)$ is $(x > 3)$:

फलन $g(x)$ होगा $(x > 3)$:

(A) $\frac{3}{(x-3)^3}$ (B) $\frac{4}{(x-3)^3}$ (C) $\frac{9}{(x-3)^3}$ (D) $\frac{27}{(x-3)^3}$

Ans. C

Sol. $f(x)=x^3$ ATQ $\frac{d}{dx}\left[f(x)\cdot g(x)\right]=\frac{df}{dx}\cdot\frac{dg}{dx}$

$\Rightarrow 3x^2\cdot g(x)+g'(x)\cdot x^3=3x^2\cdot g'(x)$

$\Rightarrow 3x^2 g(x)=\left(3x^2-x^3\right)g'(x)$

$\Rightarrow \int\frac{3}{3-x}dx=\int\frac{dg}{g}$

$\Rightarrow -3\ell n|3-x|=\ell ng+\ell nc$

$\Rightarrow c.g(x)=\frac{1}{(3-x)^3}$

$x=0$ $\quad C\cdot\left(\frac{1}{3}\right)=\frac{1}{27}\quad\Rightarrow C=\frac{1}{9}$

$x>3$ $\quad g(x)=\frac{9}{|3-x|^3}=\frac{9}{(x-3)^3}$

21. Derivative of $\{f(x-3)\,.\,g(x)\}$ with respect to x at $x=100$ is :

$x=100$ पर $\{f(x-3)\,.\,g(x)\}$ का x के सापेक्ष अवकलज होगा :

(A) 0 (B) 1 (C) – 1 (D) 2

Ans. A

Sol. $\frac{d}{dx}\{f(x-3)\cdot g(x)\}=g'(x)f(x-3)+f'(x-3)g(x)$

$\frac{d}{dx}\{f(x-3)\cdot g(x)\}=-\frac{27}{(x-3)^4}\cdot(x-3)^3+3(x-3)^2\frac{9}{(x-3)^3}$

$=0$

$\Rightarrow \frac{d}{dx}\{f(x-3)\cdot g(x)\}_{x=100}=0$

22. $\lim\limits_{x\to 0}\frac{f(x).g(x)}{x(1+g(x))}$ will be :

$\lim_{x\to 0}\frac{f(x).g(x)}{x(1+g(x))}$ होगा :

(A) 0 (B) – 1 (C) 1 (D) 2

Ans. A

Sol. $\lim_{x\to 0}\frac{f(x)\cdot g(x)}{x\left[1+g(x)\right]}=\lim_{x\to 0}\frac{x^3\frac{9}{(x-3)^3}}{x\left[1+\frac{9}{(x-3)^3}\right]}$

$\lim_{x\to 0}\frac{9x^2}{(x-3)^3+9}$

$= 0$

Paragraph for Question Nos. 23 to 24

Let $f(x)=\frac{\cos^2 x}{1+\cos x+\cos^2 x}$ and $g(x) = \lambda \tan x + (1 - \lambda) \sin x - x$, where $\lambda \in R$ and $x \in [0, \pi/2)$.

माना $f(x)=\frac{\cos^2 x}{1+\cos x+\cos^2 x}$ तथा $g(x) = \lambda \tan x + (1 - \lambda) \sin x - x$ है जहां $\lambda \in R$ तथा $x \in [0, \pi/2)$ है।

23. g'(x) equals :

g'(x) का मान होगा :

(A) $\frac{(1-\cos x)(f(x)-\lambda)}{\cos x}$ (B) $\frac{(1-\cos x)(\lambda-f(x))}{\cos x}$

(C) $\frac{(1-\cos x)(\lambda-f(x))}{f(x)}$ (D) $\frac{(1-\cos x)(\lambda-f(x))}{(f(x))^2}$

Ans. C

Sol. $f(x)=\frac{\cos^2 x}{1+\cos x+\cos^2 x}$ $\quad x\in\left[0,\frac{\pi}{2}\right)\Rightarrow f(x)>0$

$f(x)=\frac{1}{\sec^2 x+\sec x+1}$

$$g(x)=\lambda\tan x+(1-\lambda)\sin x-x$$

$$g'(x)=\lambda\sec^2 x-\lambda\cos x+\cos x-1$$

$$=\frac{\lambda(1-\cos^3 x)}{\cos^2 x}-(1-\cos x)$$

$$=(1-\cos x)\left[\frac{\lambda(1+\cos x+\cos^2 x)}{\cos^2 x}-1\right]$$

$$=(1-\cos x)\frac{(\lambda-f(x))}{f(x)}$$

24. The exhaustive set of values of 'λ' such that $g'(x)\geq 0$ for any $x\in[0,\pi/2)$:

'λ' के सम्पूर्ण मानों का समुच्चय जिसके लिए $x\in[0,\pi/2)$ में $g'(x)\geq 0$ है :

(A) $[1,\infty)$ (B) $[0,\infty)$ (C) $\left[\frac{1}{2},\infty\right)$ (D) $\left[\frac{1}{3},\infty\right)$

Ans. D

Sol. $g'(x)=\frac{(1-\cos x)(\lambda-f(x))}{f(x)}\geq 0$

$1-\cos x=+ve \qquad \Rightarrow \lambda-f(x)\geq 0$

$f(x)=+ve \qquad \lambda\geq f(x) \qquad$ _____(1)

Now $f(x)=\frac{\cos^2 x}{1+\cos x+\cos^2 x}=\frac{1}{\sec^2 x+\sec x+1}\ x\in\left[0,\frac{\pi}{2}\right)$

$\sec^2 x=t\in[1,\infty)$

$f(t)=\frac{1}{t^2+t+1}\ t\in[1,\infty)$

$f(t)\in\left(0,\frac{1}{3}\right] \qquad \because t^2+t+1\in[3,\infty)$

$t\in[1,\infty)$

Range of f(x) is $\left(0, \frac{1}{3}\right]$

$O < f(x) \le \frac{1}{3}$ _____(2)

from (1) and (2) $\lambda \ge \frac{1}{3}$ $\Rightarrow \lambda \in \left[\frac{1}{3}, \infty\right)$

MATCH THE COLUMN TYPE QUESTION(S)

25. Match the following columns :

Column-I	**Column-II**
(A) If $f'(x) = \sqrt{3x^2+6}$ and $y = f(x^3)$ then at $x = 1$, $\frac{dy}{dx}$	(P) – 2
(B) If f be a diffr. fun. such that $f(xy) = f(x) + f(y)$, $\forall\ x, y \in R$ then $f(e) + f(1/e) =$	(Q) 4
(C) If f be a twice diffr. fun such that $f''(x) = -f(x)$ and $f'(x) = g(x)$, If $h(x) = (f(x))^2 + (g(x))^2$ and $h(5) = 9$ then $h(10) = ?$	(R) 0
(D) $y = \tan^{-1}(\cot x) + \cot^{-1}(\tan x)$, then $\frac{dy}{dx}$	(S) –1
	(T) 9

निम्न कॉलम को सुमेलित कीजिए :

Column-I	**Column-II**
(A) यदि $f'(x) = \sqrt{3x^2+6}$ तथा $y = f(x^3)$ तो $x = 1$ पर $\frac{dy}{dx}$ होगा	(P) – 2
(B) यदि f एक अवकलनीय फलन है इस प्रकार कि $f(xy) = f(x) + f(y)$, $\forall\ x, y \in R$ तो $f(e) + f(1/e) =$	(Q) 4
(C) यदि f एक दो बार अवकलनीय फलन है इस प्रकार कि(R) 0 $f''(x) = -f(x)$ तथा तथा $f'(x) = g(x)$, तथा $h(x) = (f(x))^2 + (g(x))^2$ तथा $h(5) = 9$ तो $h(10) = ?$	(R) 0
(D) $y = \tan^{-1}(\cot x) + \cot^{-1}(\tan x)$, तो $\frac{dy}{dx}$	(S) –1
	(T) 9

Ans. A – T, B – R, C – T, D – P

Sol. (A) $f'(x) = \sqrt{3x^2+6}$ $y = f(x^3)$

$\frac{dy}{dx} = f'(x^3).3x^2$

$$\left.\frac{dy}{dx}\right)_{x=1} = f'(1)\cdot 3 = \sqrt{3+6}\cdot 3 = 9$$

$A \to T$

(B) $f(xy) = f(x) + f(y)$ and $x, y \in R$ and f differentiable

$\Rightarrow f(x) = \ell nx$

$$f(x) + f\left(\frac{1}{x}\right) = 0 \qquad \Rightarrow f(e) + f\left(\frac{1}{e}\right) = 0$$

$B \to R$

(C) $f''(x) = -f(x)$ $\qquad f'(x) = g(x)$

$h(x) = \left[f(x)\right]^2 + \left[g(x)\right]^2 \qquad h(5) = 9$

$h'(x) = 2f(x)\cdot f'(x) + 2g(x)\cdot g'(x)$

$= 2f(x)\cdot f'(x) + 2f'(x).f''(x)$

$= 2f(x)\cdot f'(x) - 2f'(x)\cdot f(x) = 0$

$h'(x) = 0$

$h(x) = \text{const} \quad \Rightarrow h(5) = 9 \quad \Rightarrow h(x) = 9$

$\Rightarrow h(10) = 9$

$C \to T$

(D) $y = \tan^{-1}(\cot x) + \cot^{-1}(\tan x)$

$$y = \frac{\pi}{2} - \cot^{-1}(\cot x) + \frac{\pi}{2} - \tan^{-1}(\tan x)$$

$$y = \frac{\pi}{2} - x + \frac{\pi}{2} - x$$

$y = \pi - 2x$

$$\frac{dy}{dx} = -2 \qquad \Rightarrow D \to P$$